STATS 280 W25 - Midterm 1 - Solutions

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Ex.1 (30 points)

a 8 points

Higher percentages of men smoke than women, and women tend to live longer and be healthier than men. Hence sex is a possible confounder for the relationship between smoking and health. Performing separate comparisons for men and women controls for this possible confounder. Smoking and health also differ by age group, making the age group a possible confounder. Performing separate comparisons for each age group controls for this possible confounder.

b 7 points

Subjects who recently stopped smoking probably had to stop because smoking adversely affected their health.

c 8 points

If you want to control for n binary variables, then you will need to create 2^n tables, and each table will need at least 30 subjects. Hence the total number of subjects will be at least $30 * 2^n$. If $n > 5$, then this value will be greater than the total sample size of $1,000$. You can control for a maximum of 5 variables.

d 7 points

The subjects choose whether to smoke or not, so this is not a randomized controlled experiment. It is an observational study.

Ex.2 (30 points)

a 8 points

The completion percentage will be a weighted average of the completion percentages for the distance ranges, with the weights given by the percentage of throws for the ranges. Hence the completion percentage will be $0.805 * 0.45 + 0.65 * 0.34 + 0.52 * 0.15 + 0.336 * 0.06 = 0.68$, or 68%.

b 7 points

An average quarterback who throws with the same distribution as your quarterback would have a completion percentage of $0.805 * 0.30 + 0.65 * 0.45 + 0.52 * 0.20 + 0.336 * 0.05 = 0.65$, or 65%. Your quarterback is actually less accurate than an average quarterback with the same distribution of throws as your quarterback.

c 8 points

Let the numbers in the list be $x_1 \leq x_2 \leq x_3 \leq x_4 \leq x_5 \leq x_6$. We know that $x_1 = 1$ since x_1 is the minimum number in the list, and $x_2 = 4$ since x_2 will be the 25th percentile if $x_1 \neq x_2$. The median of the list will be $x_3 = 5$ since $x_1 \neq x_2$ and $x_2 \neq x_3$. The 75th percentile if the list will be $x_4 = 8$ if $x_4 = x_5$ and will be $x_5 = 8$ if $x_4 < x_5$. The maximum value is $x_6 = 10$. There is only one number x in the list that we do not know, and we can use the mean to solve for that number. The mean will be $\frac{1+4+5+x+8+10}{6} = \frac{28+x}{6} = \frac{35}{6}$. Hence $x = 7$ and the list is $1, 4, 5, 7, 8, 10$.

d 7 points

The mean will be somewhere between 0 and 10 if all of the numbers will be between 0 and 10 and so the differences from the mean will be somewhere between -10 and 10. This means that the squared differences will be between 0 and 100, so that the average of the squared differences will be between 0 and 100. Hence the square root of these average of squared differences, the standard deviation, will be at most 10. The standard deviation can not be greater than 10.

Ex. 3 (35 points)

a 25 points

The median is the value where 50% of the area is to the left and 50% to the right. The interval from 1 to 2 contains 35% of the area and the interval from 2 to 3 10% of the area. Hence the median is in the interval from 3 to 4. We need 5% of the 15% area in the interval from 3 to 4, and so the median is $3+0.05/0.15 = 3.33$ since the values are evenly distributed.

The mean for each block is in the middle of the block since the values are evenly distributed, and so we can estimate the mean as a weighted average $1.5 * 0.35 + 2.5 * 0.10 + 3.5 * 0.15 + 4.5 * 0.25 + 5.5 * 0.15 = 3.25$.

The 25th percentile is $1 + \frac{25}{35} = 1.71$ since the values are evenly distributed within the block, and the 75th is in the block from 4 to 5 since $0.35 + 0.10 + 0.15 = 0.60 = 60\%$ of the area is to the left of 4 and 25% of the area is between 4 and 5. The 75th percentile is $4 + \frac{15}{25} = 4.6$. Hence the interquartile range is $4.6 - 1.71 = 2.89$.

b 5 points

This is the median, or 3.33.

c 5 points

This is the mean, or 3.25.

Ex.4 (35 points)

a 8 points

Converting to $z - scores$ gives $\frac{42-43}{3} = -0.33$ and $\frac{47-43}{3} = 1.33$. The table gives the area from 0 to 1.33 as approximately $0.82/2 = 0.41$ and the area from -0.33 to 0 as approximately $0.26/2 = 0.13$. Hence the percentage between 42 and 47 inches will be approximately $0.41 + 0.13 = 0.54 = 54\%$.

b 8 points

Converting to standard units gives $(48 - 43)/3 = 1.67$. The area in the right tail tail will be approximately $(1 - 0.905)/2 = 0.05 = 5\%$.

c 8 points

The 15th percentile corresponds to 70% of the area in the middle, which corresponds roughly to $z = -1.04$ Hence the height will be $42 - 1.04 * 3 = 38.88$ inches.

d 5 points

Markov's inequality gives a maximum percentage of $43/48 = 90\%$.

e 6 points

Converting to standard units gives $(47.5 - 43)/3 = 1.5$ and $(38.5 - 43)/3 = -1.5$. Chebyshev's inequality gives a maximum of $1/1.5^2 = 44\%$.

Ex.5 (30 points)

a 8 points

We know that $corr = 0.2 = \frac{\overline{xy} - \bar{x}\bar{y}}{SD(x)SD(y)} = \frac{\overline{xy} - 43*40}{3*10}$. Solving gives $\overline{xy} = 0.2 * 3 * 10 + 43 * 40 = 1,726$, and so the sum is $1,726 * 100 = 172,600$.

b 8 points

We know that $corr = 0.2 = \frac{\overline{xy} - \bar{x}\bar{y}}{SD(x)SD(y)} = \frac{\overline{xy} - 55*70}{3*15}$. Solving gives $\overline{xy} = 0.2 * 3 * 15 + 55 * 70 = 3,859$, and so the sum is $3,859 * 100 = 385,900$.

c 8 points

We know that $corr = \frac{\overline{xy} - \bar{x}\bar{y}}{SD(x)SD(y)}$. Combining parts a. and b. gives $\overline{xy} = \frac{172,600+385,900}{100+100} = 2,792.5$. We can now calculate $corr = \frac{2,792.5-49*55}{6.7*19.7} = 0.74$

d 6 points

There is a noticeable positive relationship, and so the correlation must be 0.7.

Ex.6 (40 points)

a 10 points

Converting to standard units gives $\frac{13-10}{2} = 1.5$. The regression method predicts a length that is $1.5*0.5 = 0.75$ standard deviations above average. Hence the predicted length will be $28 + 0.75 * 3 = 30.25in$

b 10 points

The slope will be $0.5 * 3/2 = 0.75$ and the line will go through the point of averages $(10, 28)$. Hence the equation of the line will be $y - 28 = 0.75(x - 10)$, or $y = 0.75x + 20.5$.

c 12 points

The 25th and 75th percentiles will correspond to 50% of the area in the middle, and so the 25th percentile will be roughly -0.67 and the 75th percentile 0.67 from the table in the back of the book. This question is considering lengths in a vertical strip corresponding to $x = 13$, and so we will want to use the root mean squared error, $\sqrt{1 - 0.5^2} * 3 = 2.6in$, for the standard deviation. Hence the interquartile range will be $2.6 * (0.67 - (-0.67)) = 3.48in$

d 8 points

The 25th percentile corresponds to $z = -0.67$ from part c. This gives a weight of $10 - 0.67 * 2 = 8.66lb$. We can now use the regression equation in part b. to get a predicted value of $0.75 * 8.66 + 20.5 = 27in$